

Statistical Language Model:-

Finding the probability distribution over a word sequence.

Joint probability

$$P(w) = P(w_1, w_2, w_3, w_4, w_5)$$

Conditional probability

$$P(w_5 | w_1, w_2, w_3, w_4)$$

Formulae:-

$$P(x|y) = \frac{P(x, y)}{P(y)} \Rightarrow P(x, y) = P(x|y) \cdot P(y)$$

$$P(x, y, z) = P(x) P(y|x) P(z|x, y).$$

Ex: He is Smart
 x y z

$$P(\text{He}) P(\text{is} | \text{He}) P(\text{Smart} | \text{He, is})$$

Numericals:-

① Corpus

<S> I am a human </S>

<S> I am not a stone </S>

<S> I I live in Mumbai </S>

Check the probability of

<S> I I am not </S> using bigram

$$\Rightarrow P(I \text{ I am not})$$

$$= P(I|\langle s \rangle) P(I|I) P(\text{am}|I) P(\text{not}|\text{am}) P(\text{Is}|\text{not})$$

* Next is checking the probability for all that is done by count, to do that it will be written in reverse.

$$= \frac{c(\langle s \rangle | I)}{c(\langle s \rangle)} \cdot \frac{c(I | I)}{c(I)} \cdot \frac{c(I | \text{am})}{c(I)} \cdot \frac{c(\text{am} | \text{not})}{c(\text{am})} \cdot \frac{c(\text{not} | \text{Is})}{c(\text{not})}$$

* Count the similar words present from the corpus. and note it down.

$$= \frac{3}{3} \cdot \frac{1}{4} \cdot \frac{2}{4} \cdot \frac{1}{2} \cdot \frac{0}{1}$$

$$= 0$$

② Consider the following training data

$\langle s \rangle$ I am Jack $\langle Is \rangle$

$\langle s \rangle$ Jack I am $\langle Is \rangle$

$\langle s \rangle$ Jack I like $\langle Is \rangle$

$\langle s \rangle$ Jack I do like $\langle Is \rangle$

$\langle s \rangle$ do I like Jack $\langle Is \rangle$

Assume that we use a bigram Language model based on the data

What is the probable next word predicted by the model?

1. $\langle s \rangle$ Jack ...

2. $\langle s \rangle$ Jack I do

3. $\langle s \rangle$ Jack I am Jack ...

4. $\langle s \rangle$ do I like

$$P(I|\langle s \rangle) = \frac{c(\langle s \rangle | I)}{c(\langle s \rangle)} = \frac{1}{5}$$

$$P(\text{Jack}|\langle s \rangle) = \frac{c(\langle s \rangle | \text{Jack})}{c(\langle s \rangle)} = \frac{3}{5}$$

$$P(\text{do}|\langle s \rangle) = \frac{c(\langle s \rangle | \text{do})}{c(\langle s \rangle)} = \frac{1}{5}$$

$$P(\text{am}|I) = \frac{c(I | \text{am})}{c(I)} = \frac{2}{5}$$

$$P(\text{like}|I) = \frac{c(I | \text{like})}{c(I)} = \frac{2}{5}$$

$$P(\text{do}|I) = \frac{c(I | \text{do})}{c(I)} = \frac{1}{5}$$

$$P(\langle s \rangle | \text{Jack}) = \frac{c(\text{Jack} | \langle s \rangle)}{c(\text{Jack})} = \frac{2}{5}$$

$$P(\langle s \rangle | \text{like}) = \frac{c(\text{like} | \langle s \rangle)}{c(\text{like})} = \frac{2}{3}$$

$$P(\langle s \rangle | \text{am}) = \frac{c(\text{am} | \langle s \rangle)}{c(\text{am})} = \frac{1}{2}$$

$$P(I | \text{Jack}) = \frac{c(\text{Jack} | I)}{c(\text{Jack})} = \frac{3}{5}$$

$$P(\text{like} | \text{do}) = \frac{c(\text{do} | \text{like})}{c(\text{do})} = \frac{1}{2}$$

$$P(\text{Jack} | \text{like}) = \frac{c(\text{like} | \text{Jack})}{c(\text{like})}$$

$$= \frac{1}{3}$$

$$P(\text{Jack} | \text{am}) = \frac{c(\text{am} | \text{Jack})}{c(\text{am})}$$

$$= \frac{1}{2}$$

Let us check the question:-

a) $\langle s \rangle$ Jack ... I ...

Now in the probability we found, check where the Jack is coming in 2nd position.

So found out, $P(\langle s \rangle | \text{Jack}) = \frac{2}{5}$

$$P(I | \text{Jack}) = \frac{3}{5}$$

b) Jack I do ... I & like will have the same probability

c) Jack I am Jack ... I ...

d) do I like $\langle s \rangle$.

③

"Corpus"

The girl bought a chocolate

Q: The girl ...

The boy ate the chocolate

The girl bought a toy

The girl played with a toy.

$$P(\text{bought} | \text{The girl}) = \frac{c(\text{The girl} | \text{bought})}{c(\text{The girl})} = \frac{2}{3} = 0.67$$

$$P(\text{Played} | \text{The girl}) = \frac{c(\text{The girl} | \text{played})}{c(\text{The girl})} = \frac{1}{3} = 0.33.$$